

Lecture 6 : Continuity

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1. Continuity

Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let $c \in D$. f is said to be **continuous** at c , if given any neighborhood V of $f(c)$, there exists a neighborhood W of c , such that for all $x \in W \cap D$, $f(x) \in V$.

2. Equivalent definition of continuity

Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let $c \in D$. f is said to be **continuous** at c , if for a preassigned positive number ϵ , there exists a positive number δ , such that

$$|f(x) - f(c)| < \epsilon, \text{ for all } x \in N(c, \delta) \cap D.$$

3. Continuity on a set

Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let $A \subset D$, then f is said to be **continuous** on A , if f is continuous at every point of A .

4. Continuity on an interval

Let I be an interval and $f : I \rightarrow \mathbb{R}$ be a function. Let c be an interior point of I . Then, f is said to be continuous at c , if for a preassigned positive number ϵ , there exists a positive number δ , such that

$$|f(x) - f(c)| < \epsilon, \text{ for all } x \in N(c, \delta) \cap I.$$

Equivalently, f is said to be continuous at c , if for a preassigned positive number ϵ , there exists a positive number δ , such that

$$|f(c + h) - f(c)| < \epsilon, \text{ for all } h \text{ satisfying } |h| < \delta \text{ and } c + h \in I.$$

Examples

1. Let $k \in \mathbb{R}$. Prove that the function f defined by $f(x) = k$, for all $x \in \mathbb{R}$ is continuous on $\mathbb{R}$.

Solution : Let $c \in \mathbb{R}$. Then, $|f(x) - f(c)| = |k - k| = 0$, for all $x \in \mathbb{R}$.

Let us choose $\epsilon > 0$. Then, $|f(x) - f(c)| < \epsilon$ for all $x \in N(c, \delta)$ for every positive δ . This proves that f is continuous at c .

Since, c is arbitrary, f is continuous on $\mathbb{R}$.

2. Show that function f define by $f(x) = [x]$, $x \in \mathbb{R}$ is discontinuous on $\mathbb{Z}$ ($\mathbb{Z}$ is the set of all integers).

Solution : Let $c \in \mathbb{Z}$. Then, $|f(x) - f(c)| = |f(x) - c|$ for all $x \in \mathbb{R}$ for all $x \in \mathbb{R}$.

From the definition of function f , we have $|f(x) - c| \geq 1$, when $x < c$.

Let us choose $\epsilon = \frac{1}{2}$. Then,

$|f(x) - c| > \epsilon$, when $x \in (c - \delta, c)$ for every positive number δ and $(c - \delta, c) \subset (c - \delta, c + \delta)$.

Therefore, $|f(x) - f(c)| > \epsilon$ for some $x \in N(c, \delta)$ for all $\delta > 0$. Since, c is arbitrary, hence f is discontinuous on $\mathbb{Z}$.

5. Continuity from left

Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let $c \in D$. f is said to be **continuous from left** at c , if $\lim_{x \rightarrow c^-} f(x)$ exists and is equal to $f(c)$.

6. Continuity from right

Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let $c \in D$. f is said to be **continuous from right** at c , if $\lim_{x \rightarrow c^+} f(x)$ exists and is equal to $f(c)$.

Note: A function f , continuous at c , if $\lim_{x \rightarrow c^-} f(x)$ and $\lim_{x \rightarrow c^+} f(x)$, both exist, finite and

$$\lim_{x \rightarrow c^-} f(x) = f(c) = \lim_{x \rightarrow c^+} f(x).$$

Example : Examine the continuity of the function $f(x)$ at $x = 0, 1, 2$, where

$$f(x) = \begin{cases} -x^2, & x \leq 0 \\ 5x - 4, & 0 < x \leq 1 \\ 4x^2 - 3x, & 1 < x < 2 \\ 3x + 4, & x \geq 2. \end{cases}$$

Solution : Since, we have to evaluate $\lim_{x \rightarrow c^-} f(x)$ and $\lim_{x \rightarrow c^+} f(x)$ (where $c = 0, 1, 2$). Thus, we are interested in the nature of the function f in the neighborhood of c .

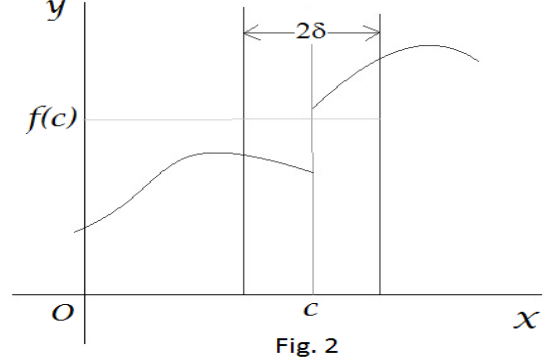
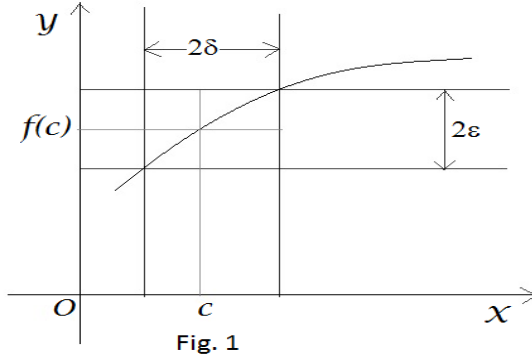
At $c = 0$, we have $\lim_{x \rightarrow 0^-} f(x) = 0$ and $\lim_{x \rightarrow 0^+} f(x) = -4$. Since, $\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$, therefore, $f(x)$ is not continuous at $x = 0$.

At $c = 1$, we have $\lim_{x \rightarrow 1^-} f(x) = 1$, $\lim_{x \rightarrow 1^+} f(x) = 1$ and $f(1) = 1$. Since, $\lim_{x \rightarrow 1^-} f(x) = f(1) = \lim_{x \rightarrow 1^+} f(x)$, therefore, $f(x)$ is continuous at $x = 1$.

At $c = 2$, we have $\lim_{x \rightarrow 2^-} f(x) = 10$, $\lim_{x \rightarrow 2^+} f(x) = 10$ and $f(2) = 10$. Since, $\lim_{x \rightarrow 2^-} f(x) = f(2) = \lim_{x \rightarrow 2^+} f(x)$, therefore, $f(x)$ is continuous at $x = 2$.

7. Geometric interpretation of continuity

When f is continuous at c , we can find an appropriate $\delta > 0$, for any given $\epsilon > 0$ and see graphically that for all $x \in N(c, \delta)$, the curve of the function f will lie in the ϵ -neighborhood of $f(c)$, i.e. $N(f(c), \epsilon) = (f(c) - \epsilon, f(c) + \epsilon)$.



The dependence of δ on ϵ is illustrated in Fig.1. We see that, when f is continuous we can find appropriate δ for any given ϵ .

This will not happen when, f is discontinuous at c (Fig.2). At such point c , however small we choose our δ , we cannot make $|f(x) - f(c)|$ arbitrarily small for all $x \in N(c, \delta)$.

Theorem 1 Let $D \subseteq \mathbb{R}$ and function f and g defined on $D \rightarrow \mathbb{R}$. Let $c \in D$ and function f and g are continuous at c . Then,

1. $f + g$ is continuous at c ;
2. $f - g$ is continuous at c ;
3. if $k \in \mathbb{R}$, kf is continuous at c ;
4. fg is continuous at c ;

Proof:

1. We have,

$$\begin{aligned} |(f + g)(x) - ((f + g)(c))| &= |f(x) + g(x) - (f(c) + g(c))| \\ &\leq |f(x) - f(c)| + |g(x) - g(c)|. \end{aligned}$$

Let us choose, $\epsilon > 0$.

Since, f is continuous at c , therefore, $\lim_{x \rightarrow c} f(x) = f(c)$ and so there exists a positive number δ_1 , such that

$$|f(x) - f(c)| < \frac{\epsilon}{2} \text{ for all } x \in N(c, \delta_1) \cap D.$$

Since, g is continuous at c , therefore, $\lim_{x \rightarrow c} g(x) = g(c)$ and so there exists a positive number δ_2 , such that

$$|g(x) - g(c)| < \frac{\epsilon}{2} \text{ for all } x \in N(c, \delta_2) \cap D.$$

Let $\delta = \min\{\delta_1, \delta_2\}$.

Then, $|f(x) - f(c)| < \frac{\epsilon}{2}$ and $|g(x) - g(c)| < \frac{\epsilon}{2}$ for all $x \in N(c, \delta) \cap D$. Hence,

$$\begin{aligned} |(f+g)(x) - (f+g)(c)| &\leq |f(x) - f(c)| + |g(x) - g(c)| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \text{ for all } x \in N(c, \delta) \cap D. \end{aligned}$$

This implies, $|(f+g)(x) - (f+g)(c)| < \epsilon$ for all $x \in N(c, \delta) \cap D$.

This proves that, $\lim_{x \rightarrow c} (f+g)(x) = (f+g)(c) = (f+g)(c)$ and thus $f+g$ is continuous at c .

2. We have,

$$\begin{aligned} |(f-g)(x) - ((f-g)(c))| &= |(f(x) - g(x)) - (f(c) - g(c))| \\ &= |(f(x) - f(c)) - (g(x) - g(c))| \\ &\leq |f(x) - f(c)| + |g(x) - g(c)|. \end{aligned}$$

Let us choose, $\epsilon > 0$.

Since, f is continuous at c , therefore, $\lim_{x \rightarrow c} f(x) = f(c)$ and so there exists a positive number δ_1 , such that

$$|f(x) - f(c)| < \frac{\epsilon}{2} \text{ for all } x \in N(c, \delta_1) \cap D.$$

Since, g is continuous at c , therefore, $\lim_{x \rightarrow c} g(x) = g(c)$ and so there exists a positive number δ_2 , such that

$$|g(x) - g(c)| < \frac{\epsilon}{2} \text{ for all } x \in N(c, \delta_2) \cap D.$$

Let $\delta = \min\{\delta_1, \delta_2\}$.

Then, $|f(x) - f(c)| < \frac{\epsilon}{2}$ and $|g(x) - g(c)| < \frac{\epsilon}{2}$ for all $x \in N(c, \delta) \cap D$. Hence,

$$\begin{aligned} |(f - g)(x) - (f - g)(c)| &< |f(x) - f(c)| + |g(x) - g(c)| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \text{ for all } x \in N(c, \delta) \cap D. \end{aligned}$$

This implies, $|(f - g)(x) - (f - g)(c)| < \epsilon$ for all $x \in N(c, \delta) \cap D$.

This proves that, $\lim_{x \rightarrow c} (f - g)(x) = (f - g)(c)$ and hence, $f - g$ is continuous at c .

3. We have,

$$\begin{aligned} |kf(x) - kf(c)| &= |(kf)(x) - kf(c)| = |kf(x) - (kf)(c)| \\ &= |k||f(x) - f(c)|. \end{aligned}$$

Let us choose, $\epsilon > 0$.

Since, f is continuous at c , therefore, $\lim_{x \rightarrow c} f(x) = f(c)$ and so there exists a positive number δ , such that

$$|f(x) - f(c)| < \frac{\epsilon}{|k|} \text{ for all } x \in N(c, \delta) \cap D.$$

Hence,

$$|(kf)(x) - kf(c)| < |k| \frac{\epsilon}{|k|} \text{ for all } x \in N(c, \delta) \cap D.$$

This implies, $|(kf)(x) - kf(c)| < \epsilon$ for all $x \in N(c, \delta) \cap D$.

This proves that, $\lim_{x \rightarrow c} (kf)(x) = kf(c) = (kf)(c)$ and hence, kf is continuous at c .

4. We have,

$$\begin{aligned} |(fg)(x) - (fg)(c)| &= |(f(x)g(x)) - (f(c)g(c))| \\ &= |(f(x) - f(c))g(x) + f(c)(g(x) - g(c))|. \\ &\leq |f(x) - f(c)||g(x)| + |f(c)||g(x) - g(c)|. \end{aligned}$$

Since, g is continuous at c , therefore, $\lim_{x \rightarrow c} g(x)$ exists, and so there exists positive numbers B and δ_1 , such that

$$|g(x)| < B \text{ for all } x \in N'(c, \delta_1) \cap D.$$

Let $k = \max\{B, |f(c)|\}$. Then, $k > 0$ and

$$|(fg)(x) - (f(c)g(c))| < k \left\{ |f(x) - f(c)| + |g(x) - g(c)| \right\} \text{ for all } x \in N'(c, \delta_1) \cap D.$$

Let us choose, $\epsilon > 0$.

Since, f is continuous at c , therefore, $\lim_{x \rightarrow c} f(x) = f(c)$ and so there exists a positive number δ_2 , such that

$$|f(x) - f(c)| < \frac{\epsilon}{2k} \text{ for all } x \in N(c, \delta_2) \cap D.$$

Since, $\lim_{x \rightarrow c} g(x) = g(c)$ and so there exists a positive number δ_3 , such that

$$|g(x) - g(c)| < \frac{\epsilon}{2k} \text{ for all } x \in N(c, \delta_3) \cap D.$$

Let $\delta = \min\{\delta_1, \delta_2, \delta_3\}$.

Then, $\left| (fg)(x) - (fg)(c) \right| < k \cdot \frac{\epsilon}{2k} + k \cdot \frac{\epsilon}{2k}$ for all $x \in N(c, \delta) \cap D$.

This implies, $\left| (fg)(x) - (fg)(c) \right| < \epsilon$ for all $x \in N(c, \delta) \cap D$.

This proves that, $\lim_{x \rightarrow c} (fg)(x) = (fg)(c)$ and hence, fg is continuous at c .

Reference book : *Introduction to real analysis by S.K. Mapa.*